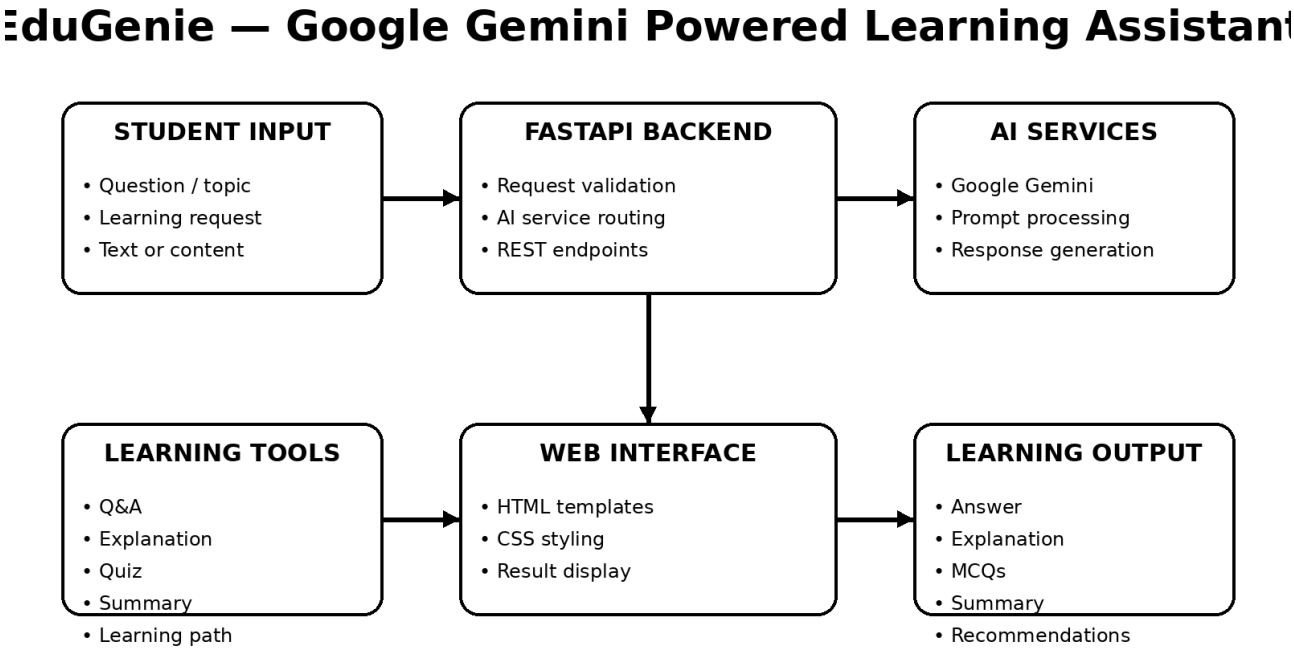
EDUGENIE

Google Gemini Powered Learning Assistant

Project Documentation



Modular AI learning workflow

An AI-powered learning assistant designed to support question answering, explanations, quizzes, summaries, and learning recommendations.

1. Project Overview

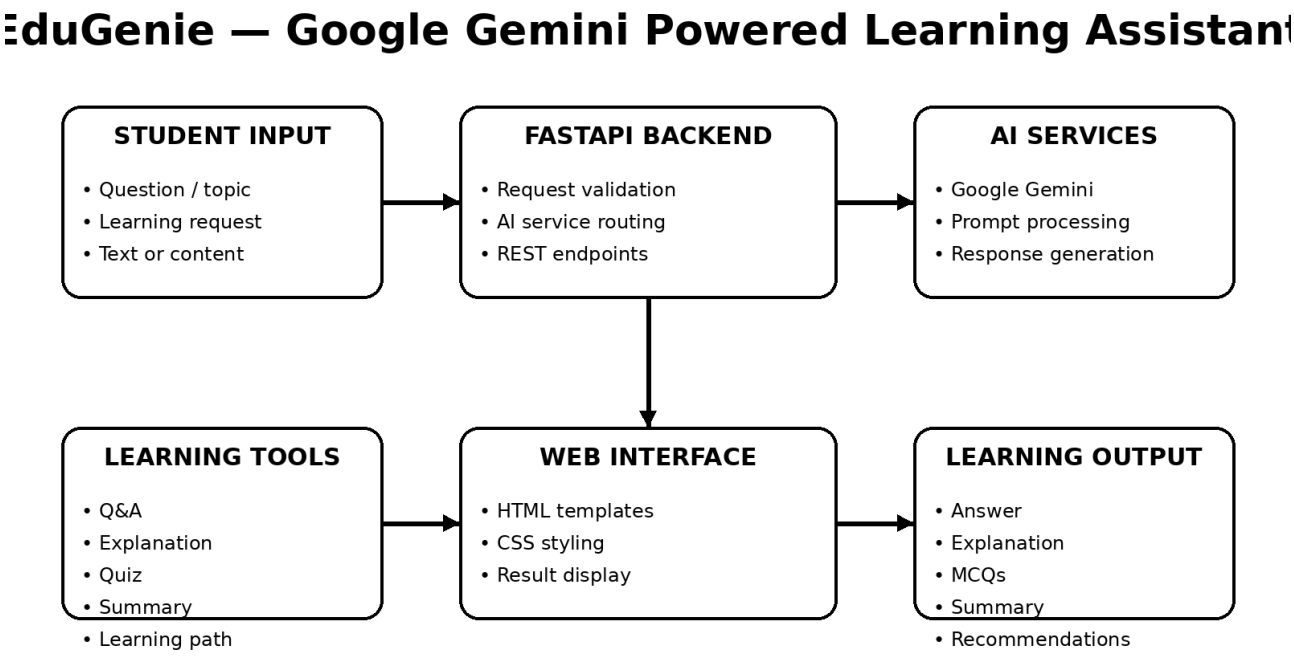
EduGenie is presented as a Google Gemini powered learning assistant. The project focuses on using generative AI to support learning tasks through a simple web application and modular backend services.

This document is organized using the supplied project-documentation reference structure: overview, setup, running instructions, web interface, API reference, operational notes, and project structure.

2. Project Objectives

- Provide AI-assisted answers to learning questions.
- Explain concepts in a learner-friendly way.
- Generate quizzes for practice.
- Summarize educational content.
- Recommend a structured learning path.

3. System Architecture



Modular AI learning workflow

4. Main Modules

Question Answering

Accepts a learner’s question and returns an AI-generated response.

Concept Explanation

Produces an explanation intended to make a topic easier to understand.

Quiz Generation

Creates practice questions from supplied learning content.

Summarization

Condenses longer educational material into a shorter form.

Learning Recommendations

Provides a structured learning path based on the requested topic or learner need.

5. Technology and Components

Component	Role
Python / FastAPI	Backend application and API layer
Jinja2 / HTML	Web interface and page rendering
CSS	Interface styling and responsive presentation
Google Gemini	Generative AI capability
API endpoints	Connect frontend requests to learning functions

6. Installation and Running

The supplied documentation reference demonstrates a Windows virtual-environment workflow. Adapt the exact dependency names and entry-point module to the EduGenie source package.

Typical steps

- Create and activate a Python virtual environment.
- Upgrade pip and install the project requirements.
- Configure the required Gemini API key in the environment configuration.
- Start the FastAPI application with Uvicorn.
- Open the local web application in a browser.
- Use the API documentation endpoint when available for testing.

7. API / Backend Workflow

The backend receives a learning request, validates the input, routes the request to the appropriate learning module and AI service, processes the generated response, and returns the result to the frontend.

Recommended test flow

- Test each learning function with a short sample input.
- Verify that valid requests return readable output.
- Check invalid input handling.
- Confirm the Gemini API key is loaded only through configuration.
- Use the interactive API documentation when available.

8. Web Interface

- Learner enters a question, topic, or educational text.
- Learner selects the required learning function.
- Frontend sends the request to the backend.
- Generated content is displayed in the result area.
- The interface should remain simple and responsive for student use.

9. Operational Notes

API credentials should be stored in environment configuration and not committed to source control. The documentation reference also emphasizes health checks, API documentation, and clear separation of application components.

10. Testing Checklist

- Question-answer response test
- Concept explanation test
- Quiz generation test
- Summary generation test
- Learning-path recommendation test
- API error/validation test
- Frontend display and responsive-layout test

11. Conclusion

EduGenie brings several common learning activities into one AI-assisted application. Its modular structure makes it possible to maintain separate learning functions while presenting them through a single student-facing interface.

12. Future Scope

- Voice-based learning
- Multilingual learning support
- Mobile application
- Learning progress analytics
- Adaptive learning recommendations
- Image/PDF-based question answering
- Teacher or administrator dashboards

Source note: This report is grounded in the uploaded EduGenie project document and uses the uploaded Project Documentation PDF as the structural reference. Where the reference PDF contains FitBuddy-specific details, those details were not presented as EduGenie facts.